Ladrilhos Automáticos

Por João Vitor Santiago Nogueira

Nos primórdios

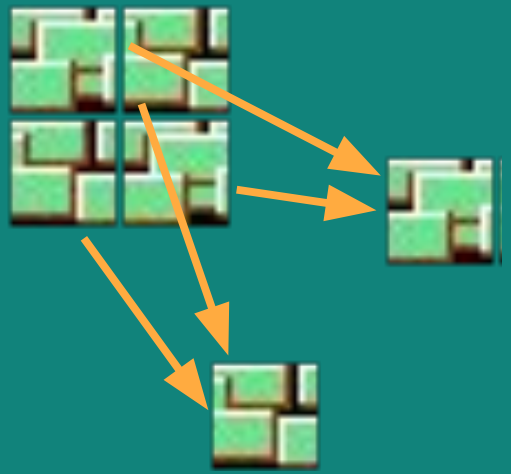
Jogos tinham pouquíssima memória.

A mesma sprite tinha que ser usada várias vezes.

Isso valia especialmente para terreno, que precisa de dezenas de blocos.







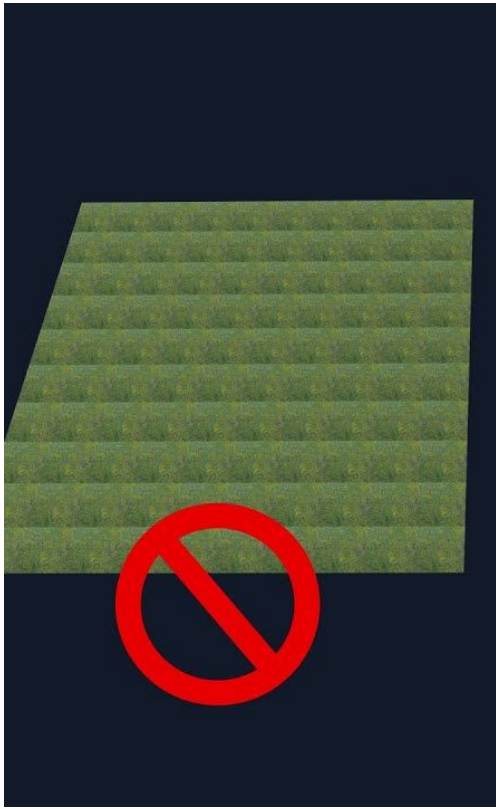
Feito de várias maneiras

Inviável guardar posições individuais.

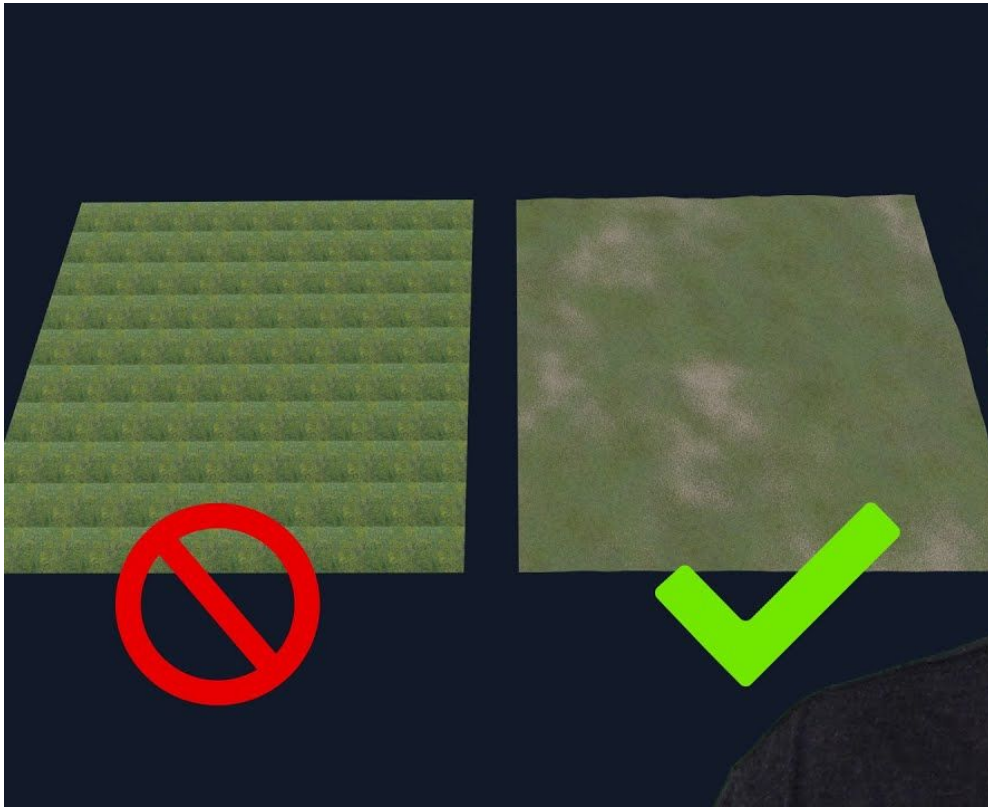
Jogos definiram regiões onde existia um tipo de objeto.

- Menos esforço colocando
- Menos memória

Só botar uma tile repetida é perceptível



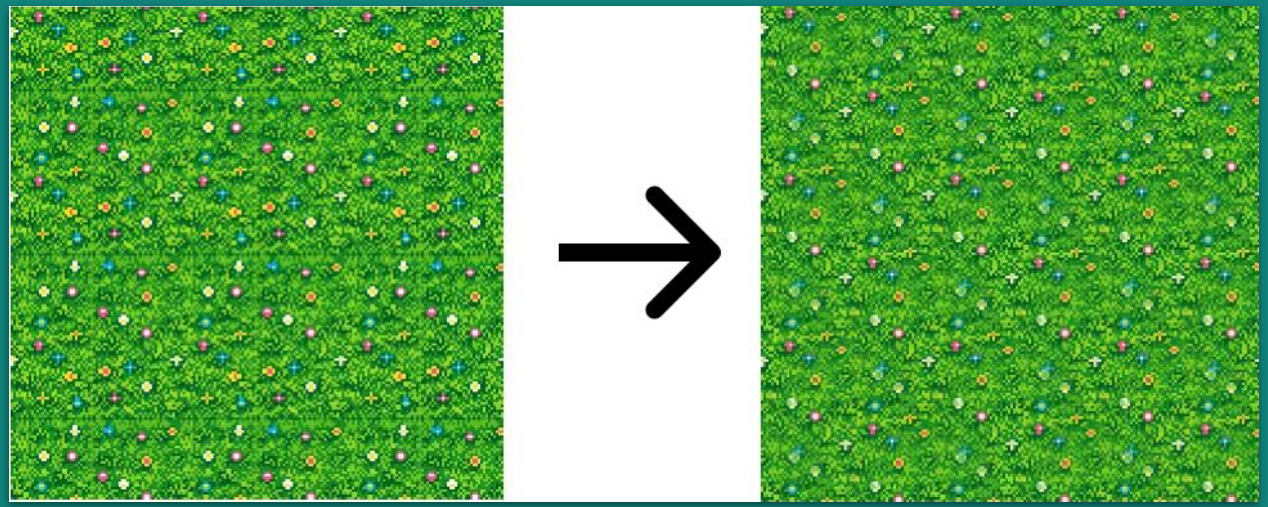
Só botar uma tile repetida é perceptível - Como suavizar?

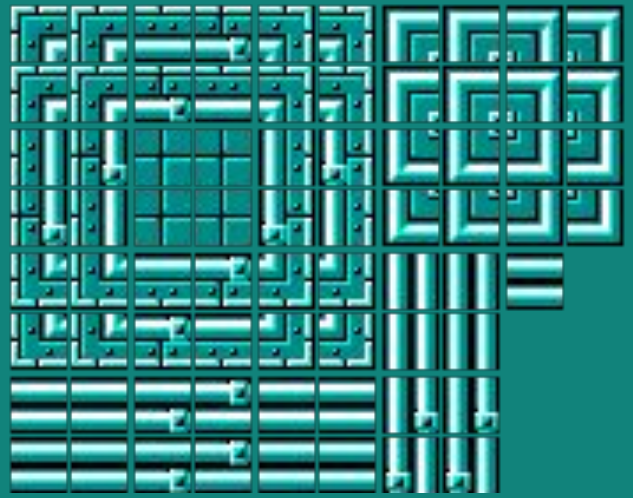


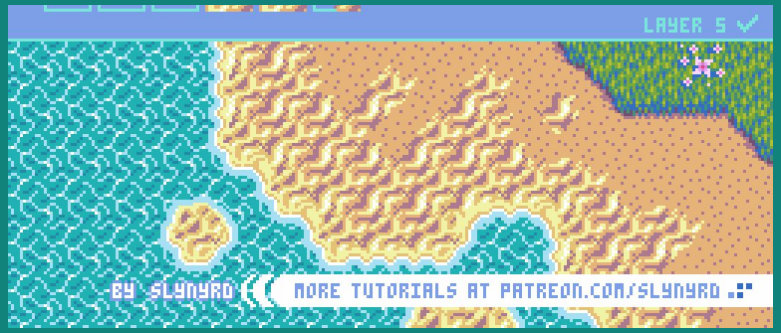
Várias maneiras

- Tiles mais espertas
- Layering
- Automatic tile placement

Tiles melhor divididas - artístico







Usar mais tiles

Seja sobrepondo, ou fazendo ou incluindo variações.

Criamos várias opções.

Botar isso na mão é trabalhoso.

Nosso editor

Não temos como botar esses macro-objetos de maneira automática.

Ainda.

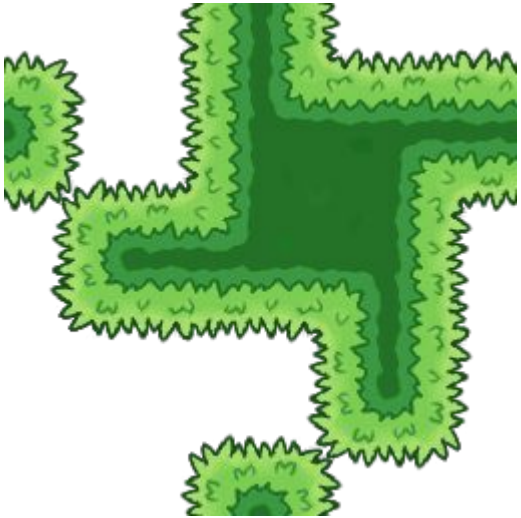


Duas abordagens

Grades

Mais usual, mais fácil

Mais limitado, mais trabalhoso



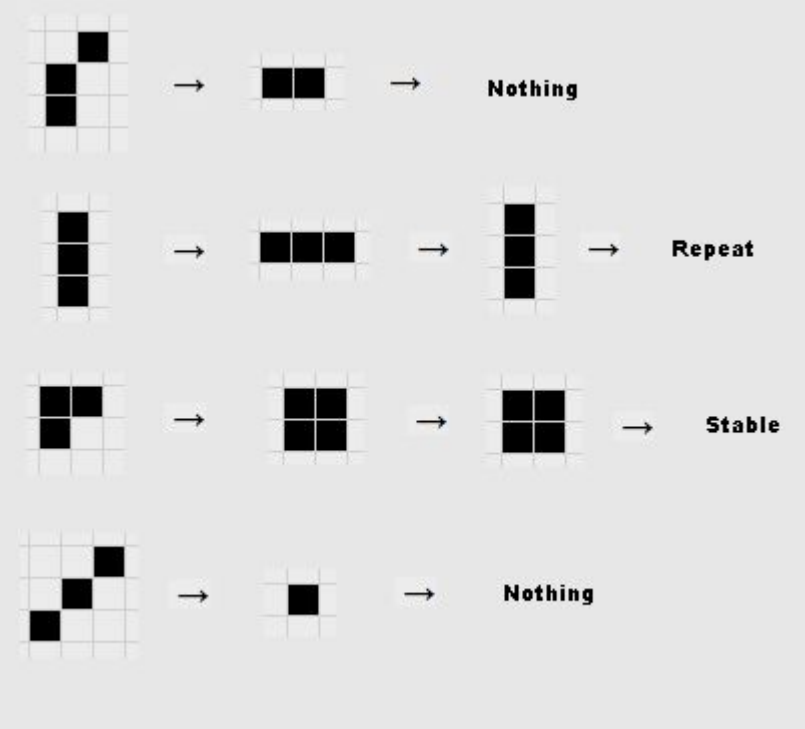
Polígonos/Curvas

Mais raro, mais complexo

Mais poderoso, mais rápido



Análogo a um autômato-celular



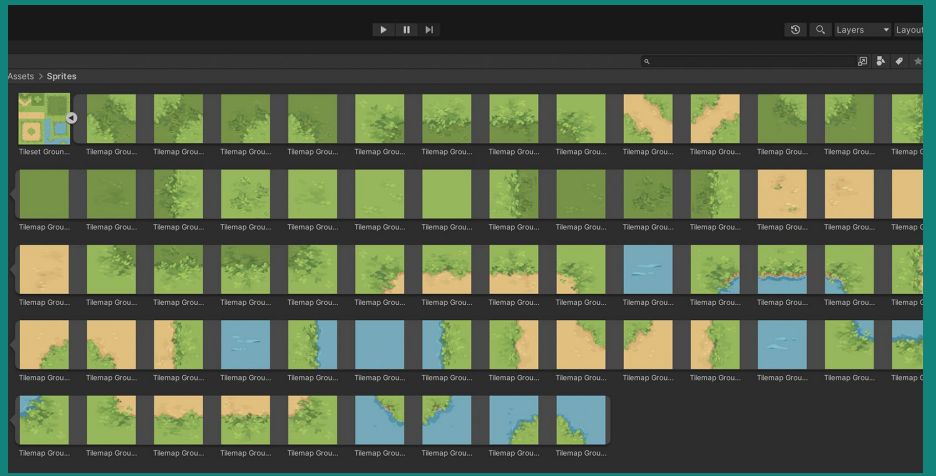


Grades

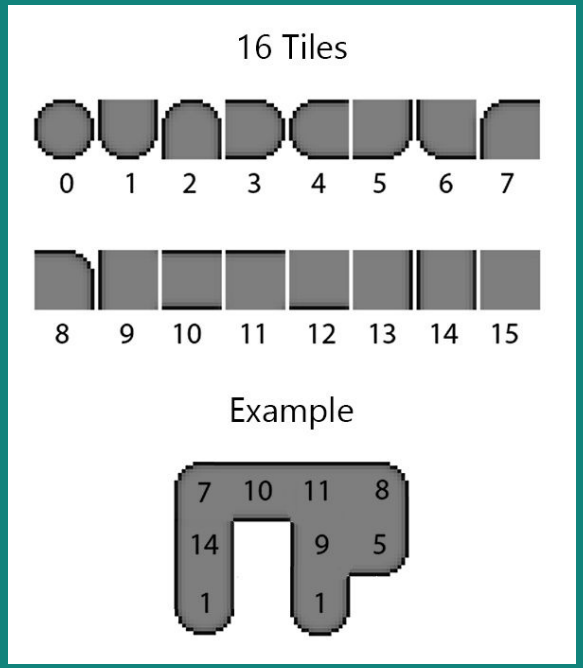
Cada célula, recebe uma sprite com base no conteúdo dela

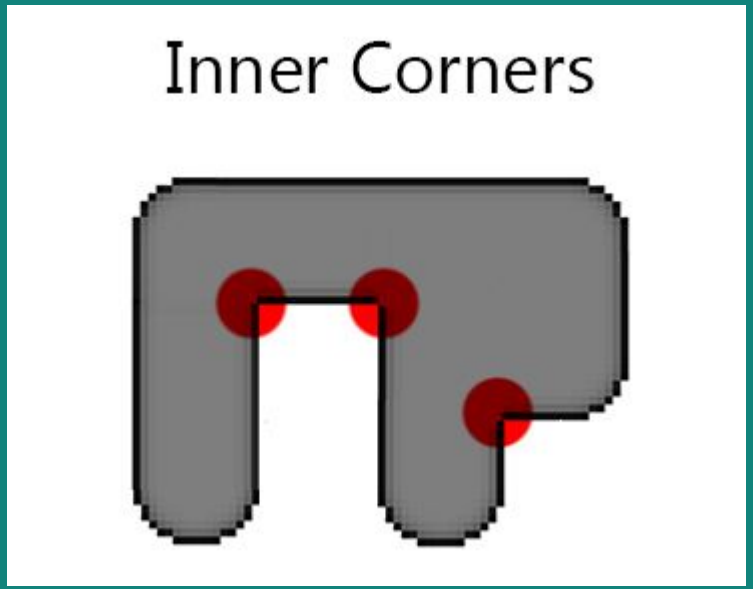
Pode reagir a celular vizinhas para se ajustar.

As vezes precisamos de células de borda/transição

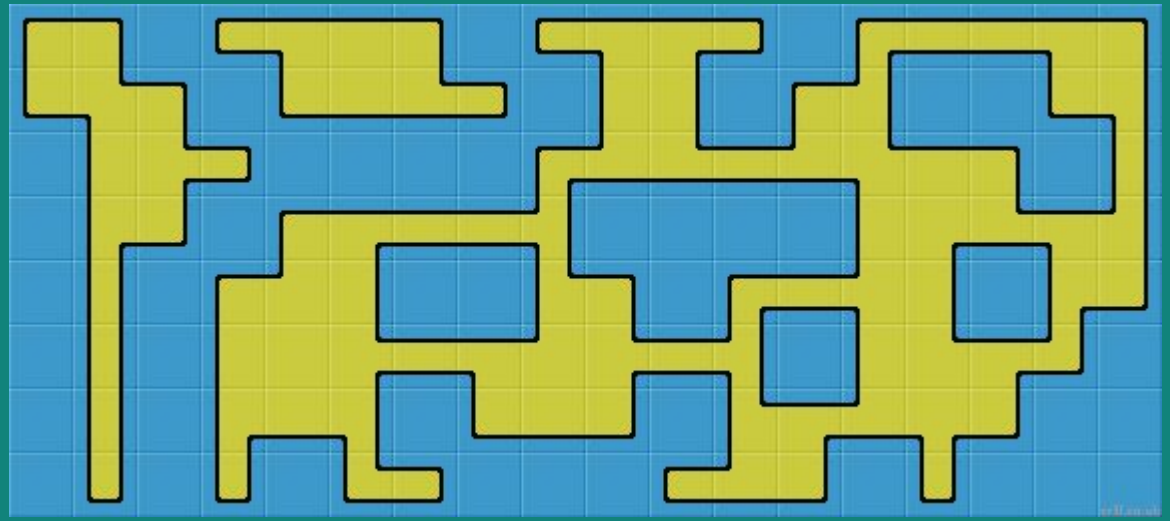


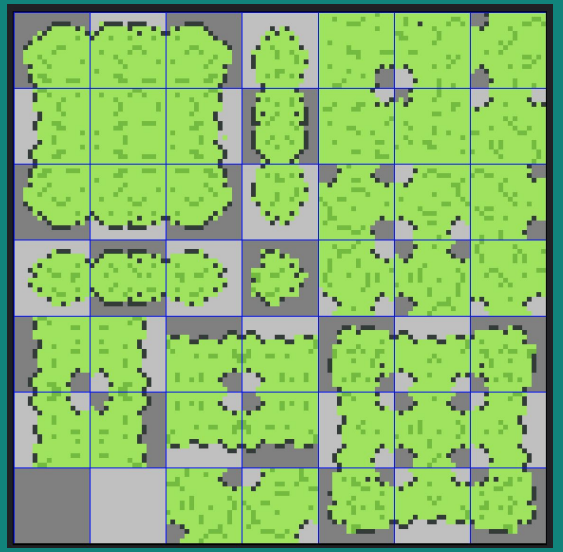
Padrão 16 - só leva em conta 4 adjacente (4^2)



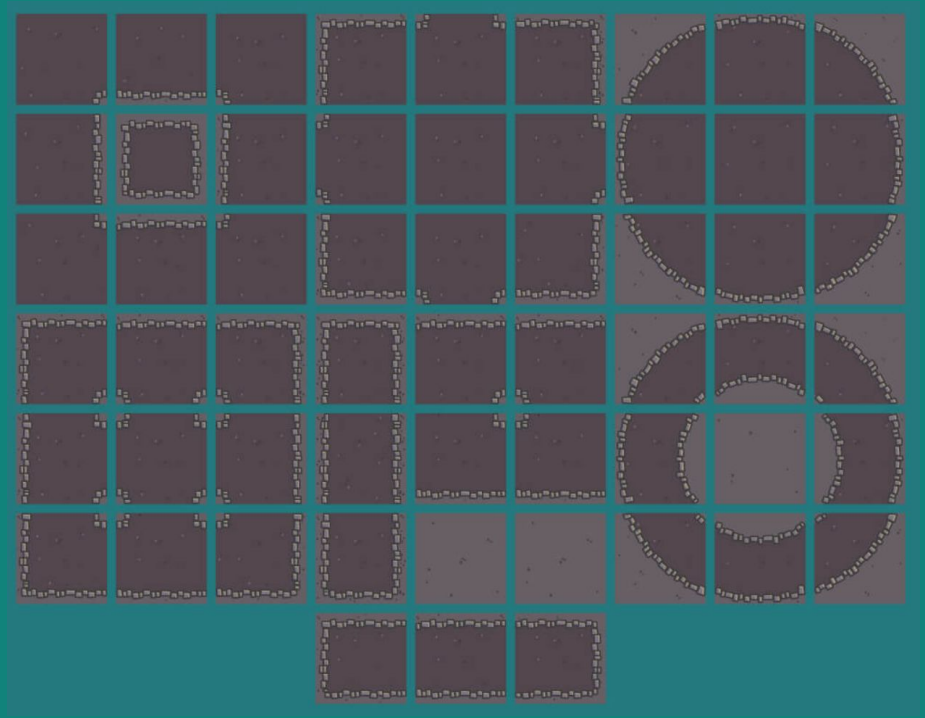


Padrão 47 - 4 adjacente (4^2) + diagonais com conexão

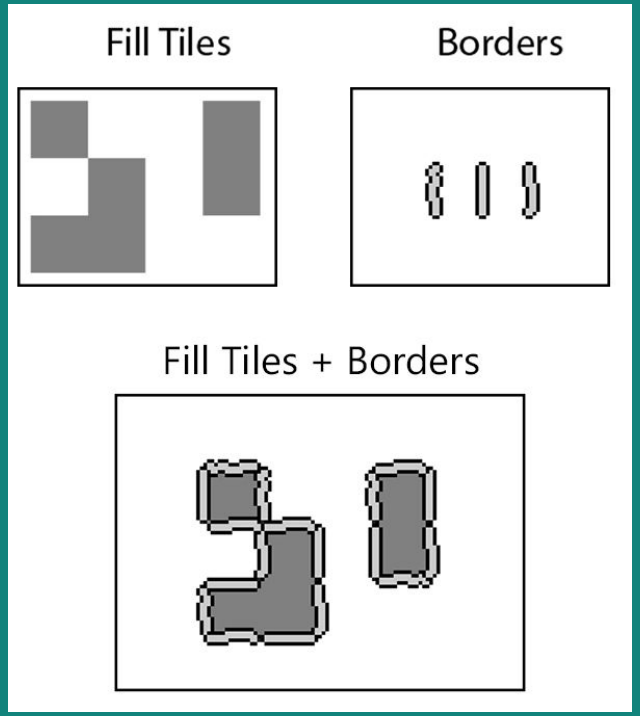


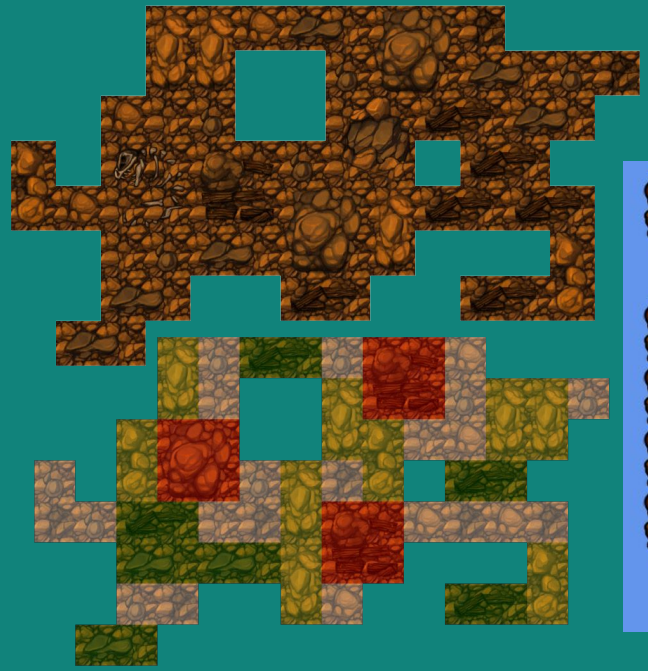


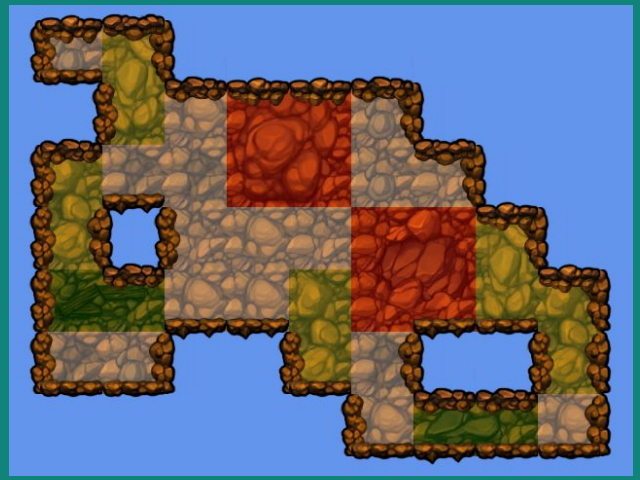
Até padrão 57



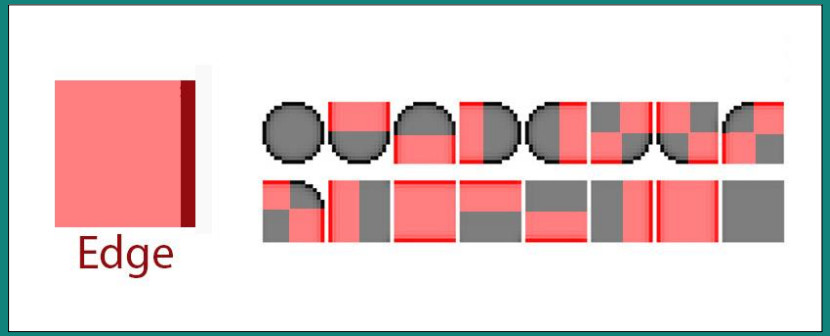
Desacoplar bordas e meios

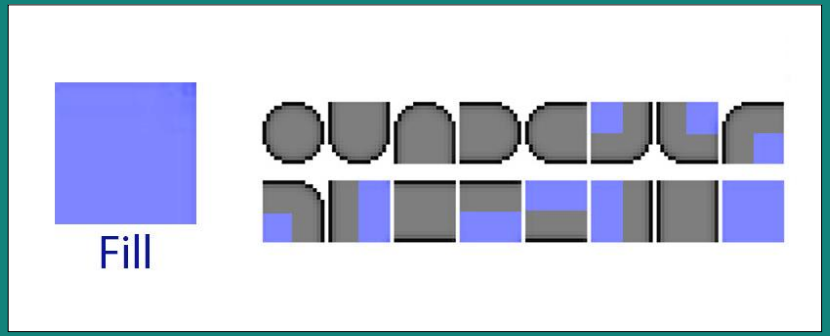


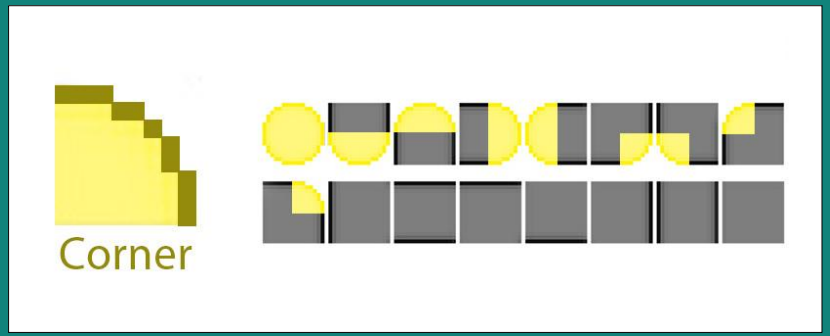


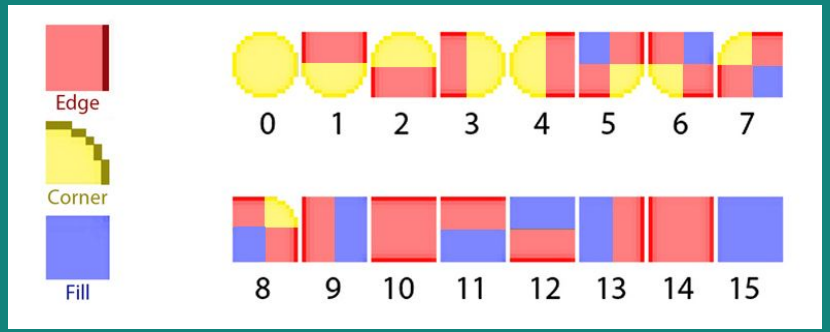


Subdivisão - Compor cada tile com sub tiles - 8-4-4

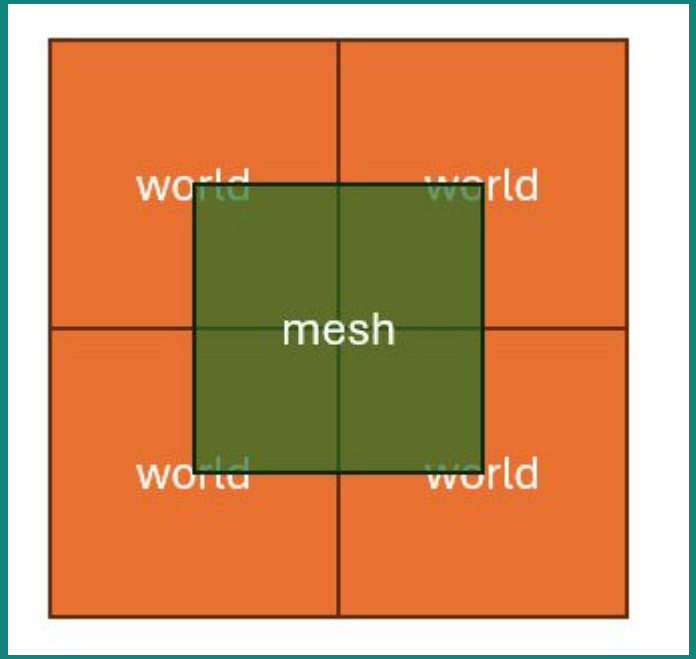


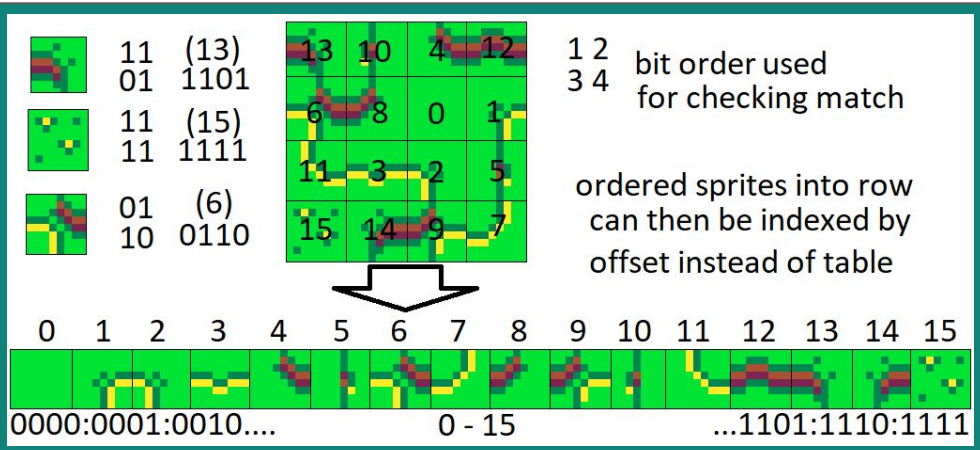


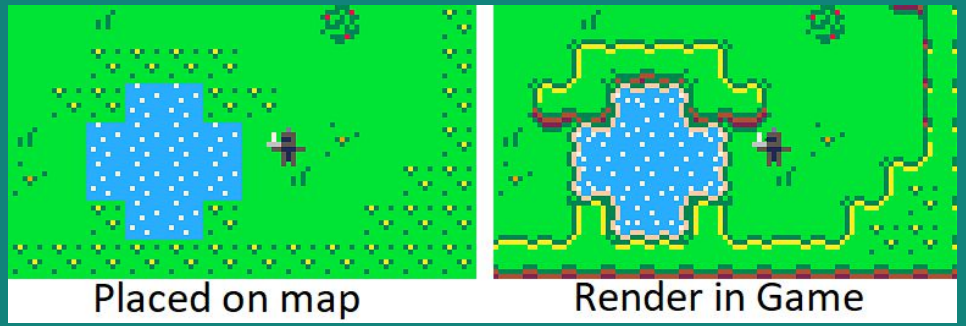




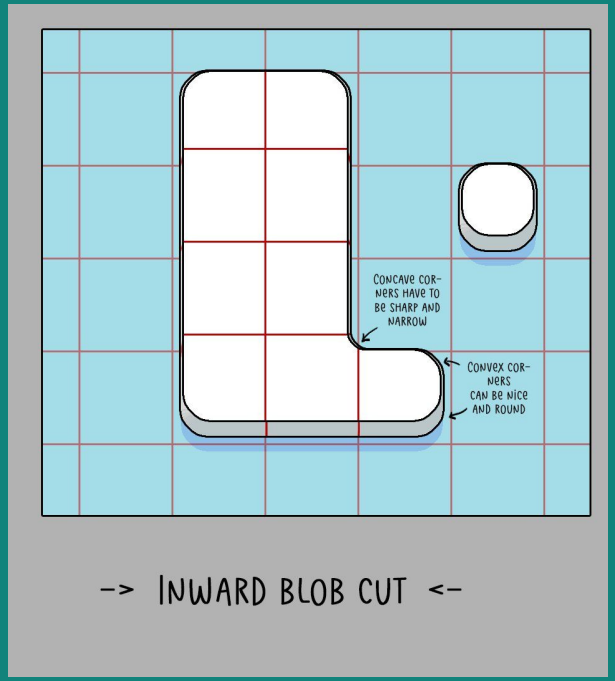
Grid dupla, desacopla - só liga para os cantos (4^2)

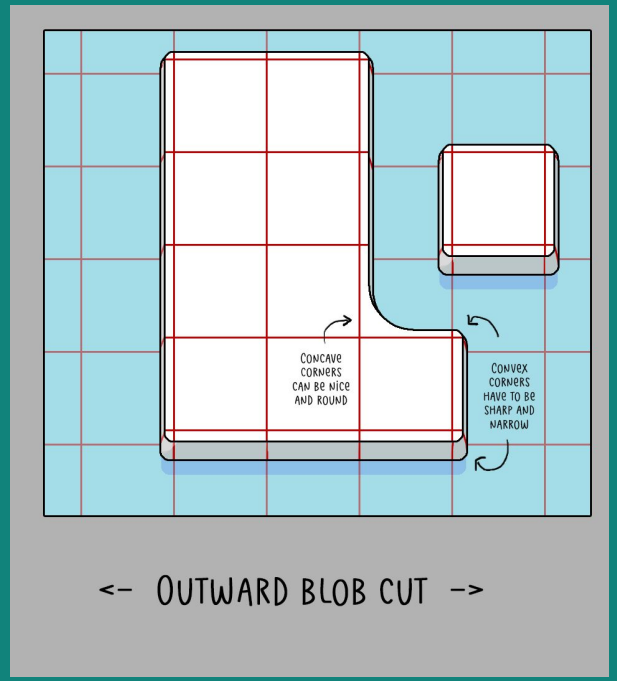


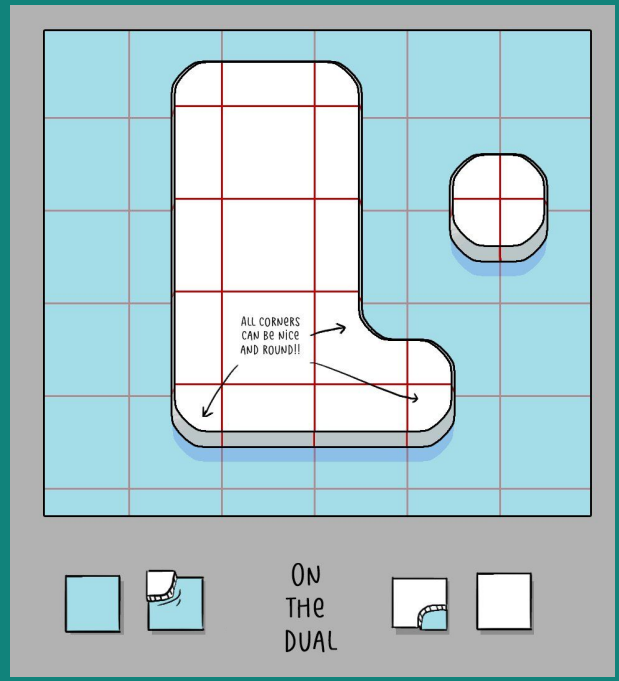




Grid dupla - permite cantos mais arredondados

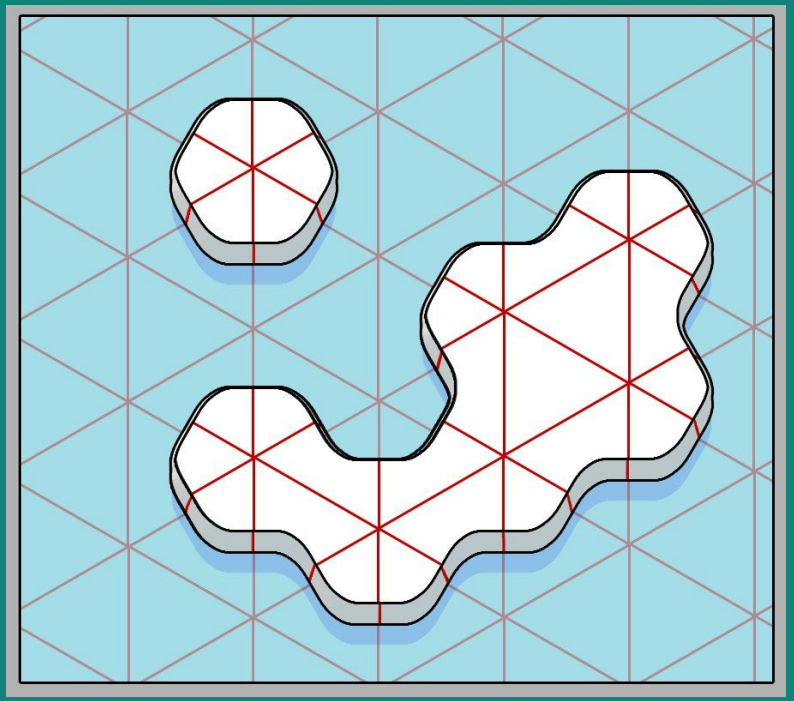






Também temos hexágonos





Extras

https://www.boristhebrave.com/permanent/24/06/cr31/stagecast/wang/blob.html

https://www.boristhebrave.com/2023/05/31/quarter-tile-autotiling/

https://www.gamedeveloper.com/programming/creating-a-dynamic-tile-system

https://github.com/Jewelots/Spelunky-Tiles

https://blog.bscotch.net/post/splattertiles-or-how-to-tile-your-game-without-all-those-tiles/

Futuramente

Marching cubes

https://www.boristhebrave.com/2021/12/29/2d-marching-cubes-with-multiple-colors/

Wave Function Collapse

https://robertheaton.com/2018/12/17/wavefunction-collapse-algorithm/

Polygonal/Curved terrain

Simulação de areia
https://www.youtube.com/watch?v=5Ka3tbbT-9E